

Algebra Interview Comparative Report

Introduction

This report compares the algebra perspectives of four individuals: one adult educator and three minor students. I explored their understanding, learning experiences, and views on algebra's purpose. A key aspect of this comparison is the significant time gap in educational experiences, with the adult interviewee having learned algebra about two decades ago, when teaching methods differed from today's. The findings highlight how abstract thinking and mathematical understanding are shaped by both time and educational context.

Interviewee Descriptions

I interviewed four individuals. Ms. Reyes, in her 30s, is an elementary inclusionary support specialist. She attended Edcouch Elsa High School, a high-poverty area school with historically low academic scores, and her algebra experience dates back about 20 years. The minor students include D.P., a 16-year-old 10th grader at Hidalgo High School; K.E., a 9th grader at Edinburg North High School; and M.W., a 13-year-old 8th grader at South Texas Preparatory Academy, a state-recognized school. M.W. shows surprisingly advanced conceptual understanding for his age.

Comparative Analysis

Conceptual Understanding of Algebra

The minor students demonstrated sophisticated abstract thinking, contrasting with Ms. Reyes's recollection of algebra as challenging. M.W. offered a philosophical view, describing algebra as using variables to "abstract complex math equations into a general format." This abstract conceptualization from a student at a state-recognized academy stood out against Ms. Reyes's memory of algebra as a difficult "first introduction to abstract concepts," an experience likely shaped by her high school and the teaching methods of her era. D.P. showed strong conceptual clarity, while K.E. focused more on procedures. The minors seemed to embrace abstraction naturally, while the adult recalled it as a difficult transition.

Purpose and Application Perspectives

The interviews showed a clear pattern in how different age groups view algebra's purpose. The minor students focused on algebra's problem-solving power and theoretical applications. M.W. saw equations as "abstract ideas that can be manipulated... to get answers to complex problems in our everyday lives." D.P. and K.E. similarly emphasized problem-solving and using variables.

In contrast, Ms. Reyes emphasized practical engagement, suggesting algebra lessons need to be "more engaging by using practical applications like budgeting,

and sports.” This difference points out a gap between abstract teaching and connecting algebra to tangible, everyday situations.

Educational Experience Patterns

Comparing current learning with past experiences revealed significant differences, influenced by time and evolving standards. The minor students described active engagement with advanced concepts like “manipulating polynomials” (M.W.), “equations, inequality’s, and Domain and range” (K.E.), and “formulas, variables, and solving for unknowns” (D.P.). These reflect modern pedagogical approaches.

Ms. Reyes, reflecting on her school experience from 20 years ago, remembered algebra as difficult and criticized approaches like “just giving them a paper and pencil.” Her focus on “PEMDAS” suggests a rules-based learning experience, common in her era and potentially impacted by her high school’s resources.

Communication and Expression Styles

Language and communication styles varied noticeably. The minor students, especially M.W. from a state-recognized academy, demonstrated impressive vocabulary and conceptual sophistication. Ms. Reyes communicated in a more casual, practical style, focusing on concrete outcomes. This difference may reflect generational changes in education, the impact of her high school’s academic environment, and the contrast between current learning versus retrospective reflection.

Key Similarities and Differences

All interviewees recognized variables as central, viewed algebra as a problem-solving tool, and understood it as building on previous math.

However, striking differences emerged: minors embraced abstract thinking while the adult recalled it as challenging, a contrast influenced by evolving teaching methods and school resources over two decades. Current students focused on theoretical applications, while the educator emphasized practical engagement. Minors used technical, sophisticated language, while the adult employed casual, practical communication.

Among the minors, M.W. showed philosophical depth, D.P. thoughtful conceptual problem-solving, and K.E. a technical, procedural focus.

Synthesis and Educational Implications

These interviews suggest important takeaways for math education. First, current students show comfort with abstract concepts that adults recall as challenging. This could indicate improved teaching, natural adaptation, or the significant impact of school resources and academic focus, as seen between Ms. Reyes’s and M.W.’s high school experiences.

Second, a distinction exists between purpose-driven (minors) and process-driven (adult) views of algebra. This suggests that distance from algebra education can lead to more critical reflection on the learning process, especially if initial environments were less supportive.

Third, the gap between technical precision and practical application highlights a curriculum opportunity. While current students grasp advanced concepts, there's a clear need to bridge this theoretical knowledge with real-world applications, as Ms. Reyes suggested. The evolution of math teaching over two decades likely plays a significant role, with modern pedagogy often emphasizing conceptual understanding and real-world relevance more explicitly.

Conclusion

This comparative analysis reveals that perspectives on algebra evolve significantly across different life stages and educational experiences. Current students demonstrate sophisticated abstract thinking and technical knowledge, while adults reflect on the learning process with practical insights. These differences are profoundly shaped by educational context, including institutional quality and pedagogical approaches over time. Successful algebra education should leverage students' conceptual strengths while incorporating practical applications, bridging abstract theory with real-world relevance, and acknowledging diverse educational backgrounds.

Complete interview transcripts follow in the appendix section.

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Interviews conducted: 4 individuals (1 adult, 3 minors)